SULIT

Fakulti Komputeran

UNIVERSITI TEKNOLOGI MALAYSIA

FINAL EXAMINATION

SEMESTER 2, 2025/2026

— SET B —

SUBJECT CODE	**:**	**SCSM2223**
SUBJECT NAME	**:**	**CROSS-PLATFORM APPLICATION DEVELOPMENT**
SECTION	**:**	**ALL SECTIONS**
TIME	**:**	**(3 HOURS)**
DATE/DAY	**:**	**7 JULY 2026 / TUESDAY**
VENUES	**:**	

INSTRUCTIONS TO THE STUDENTS:

- This exam consists of **THREE (3)** sections with a total of 100 marks: Section A — Multiple-Choice Questions (20 marks), Section B — Structured Questions (45 marks), and Section C — Code Application Questions (35 marks).
- **Answer ALL questions** in the spaces provided in this question booklet.
- You have **THREE (3) HOURS** to complete this exam.
- All code answers must be written using the syntax taught in this course and must be syntactically correct. Comment statements are **NOT** required.
- This is a **CLOSED BOOK** exam. No electronic devices are permitted.

NAME	
MATRIC NO.	
SECTION	
LECTURER	

(This question booklet consists of 16 printed pages including this page)

Mock examination paper (Set B) generated for personal revision practice — not an official UTM examination paper.

SECTION A: MULTIPLE-CHOICE QUESTIONS [20 MARKS]

(INSTRUCTION: This section consists of TWENTY (20) questions. Each question carries ONE (1) mark. Circle the letter of the BEST answer in this booklet.)

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1. In the 3-tier web architecture, the middle tier (Application Tier) is responsible for:

   A. rendering HTML, CSS and JavaScript in the browser
   B. executing the business logic on the web server
   C. storing and managing the application data
   D. converting bytecode so it can run on any operating system

2. Which of the following is a characteristic of a Multi-Page Application (MPA)?

   A. The server generates a completely new HTML page for each user request
   B. The page loads once and is updated dynamically by JavaScript
   C. The backend only provides JSON data through an API
   D. Navigation is handled on the client by a router

3. In a hybrid mobile application built from a web project, the HTML, CSS and JavaScript are executed inside:

   A. the Java Virtual Machine
   B. the Android compiler
   C. a MySQL container
   D. a WebView container provided by the native shell

4. The CSS rule `box-sizing: border-box` means that:

   A. the element's border is removed from the layout
   B. margins are included inside the element's width
   C. padding and border are included inside the element's declared width
   D. the element becomes an inline box

5. In `grid-template-columns: repeat(auto-fit, minmax(260px, 1fr))`, the effect of `auto-fit` compared with `auto-fill` is that:

   A. empty tracks collapse, so the items stretch to fill the full row width
   B. empty tracks remain, leaving visible gaps at the end of the row
   C. the number of columns is always fixed at three
   D. the grid switches automatically to Flexbox on small screens

6. The BOM dialog `confirm("Delete this note?")` returns:

   A. always `undefined`
   B. the text typed by the user, or `null`
   C. `true` if the user clicks OK, or `false` if the user cancels
   D. the number of times the dialog has been shown

7. A render function `renderNotes()` runs every time the list changes. Why must `addEventListener` NOT be called inside `renderNotes()`?

   A. event listeners are only allowed inside init() by the browser
   B. a duplicate listener would be attached on every render, so one click would fire the handler multiple times
   C. addEventListener does not work on dynamically created elements
   D. it would prevent the default action of the click event

8. AJAX is best described as:

   A. a JavaScript library for simplifying DOM manipulation
   B. a text-based data format for storing and exchanging data
   C. a protocol that replaces HTTP for real-time communication
   D. a technique that lets a web page communicate with a server in the background without reloading the entire page

9. A logged-in user with a valid token tries to delete a resource that belongs to ANOTHER user, and the server refuses. The MOST appropriate status code is:

   A. 400 Bad Request
   B. 401 Unauthorized
   C. 403 Forbidden
   D. 404 Not Found

10. The purpose of jQuery's `$(document).ready(function() { … })` is to:

    A. run the code only after the DOM is fully loaded
    B. reload the document whenever data changes
    C. convert the document into a JSON object
    D. attach a click listener to the whole document

11. Deeply nested callbacks that make code hard to read and maintain ("callback hell") are PRIMARILY solved by:

    A. using synchronous XMLHttpRequest calls instead
    B. using Promises and chaining .then() (or async/await)
    C. moving all code into a single callback function
    D. polling the server with setInterval

12. In a Promise chain, the `.finally()` handler runs:

    A. only when the Promise is fulfilled
    B. only when the Promise is rejected
    C. only if no .catch() is present
    D. in every case, whether the Promise is fulfilled or rejected

13. Compared with a method, a Vue `computed` property:

    A. must always receive an argument
    B. runs again every time the component re-renders, without exception
    C. is cached and only re-runs when one of its reactive dependencies changes
    D. can only return strings

14. In `<form @submit.prevent="save">`, the `.prevent` modifier:

    A. stops the event from bubbling up to parent elements
    B. calls `event.preventDefault()`, so the browser does not reload the page on submit
    C. makes the handler run only once
    D. validates the form fields automatically

15. In Vue Router, the `<router-view />` component is:

    A. the placeholder where the component matched by the current route is rendered
    B. a link that navigates to another route when clicked
    C. the store that holds shared state between pages
    D. a directive that hides an element when the route changes

16. In `main.js`, which ordering rule must be respected when wiring a Vue application?

    A. `app.use(router)` must come before `createApp()`
    B. `createPinia()` must be installed before the router or the app crashes
    C. plugins must be installed in alphabetical order
    D. `app.mount('#app')` must be called LAST, after all plugins are installed

17. A client wants to change ONLY the gpa field of student 3, leaving the other fields untouched. The MOST appropriate request is:

    A. PUT /students/3 with only the gpa field in the body
    B. PATCH /students/3 with only the gpa field in the body
    C. POST /students/3 with the full object in the body
    D. DELETE /students/3 followed by POST /students

18. In Slim 4, the PSR-7 response object is immutable. This means `withHeader()` and `withStatus()`:

    A. return a NEW response object, so the chained result must be returned from the route
    B. modify the existing response object in place
    C. can only be called once per application
    D. must be called before getBody()->write() or the body is lost

19. Setting `PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION` on a PDO connection makes PDO:

    A. silently ignore all database errors
    B. return false from every failed query without any error information
    C. automatically retry failed queries three times
    D. throw a PDOException whenever a database error occurs

20. Which pair correctly matches the two security questions answered by authentication (AuthN) and authorization (AuthZ)?

A. AuthN: “What can you do?” — AuthZ: “Who are you?”

B. AuthN: “Is the connection encrypted?” — AuthZ: “Is the token signed?”

C. AuthN: “Who are you?” — AuthZ: “What can you do?”

D. AuthN: “Where are you?” — AuthZ: “When did you log in?”

SECTION B: STRUCTURED QUESTIONS [45 MARKS]

(INSTRUCTION: This section consists of FOUR (4) questions. Answer ALL questions in the spaces provided.)

QUESTION 1 **[8 MARKS]**

Web Application Architecture: MPA vs SPA

a) Compare how a *Multi-Page Application (MPA)* and a *Single-Page Application (SPA)* each handle a user action (for example, clicking "View my tasks"). Your answer must explain which side performs the rendering in each model and what the server returns in each model.

(4 marks)

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b) State TWO situations in which an MPA is the more suitable choice.

(2 marks)

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c) Explain why the SPA architecture is what makes hybrid mobile development (e.g., with Capacitor) possible.

(2 marks)

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QUESTION 2 **[10 MARKS]**

Asynchronous JavaScript

a) Define *"callback hell"* and explain how Promises solve this problem.

(3 marks)

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b) A student wrote the function in Figure 1 to fetch a book and print its title, but the code contains **THREE (3) errors**.

```
function getBook() {
  try {
    const response = await fetch("https://api.example.com/books/7");
    if (!response.ok) {
      throw new Error(`HTTP Error! Status: ${response.status}`);
    }
    const book = response.json();
    console.log(book.title);
  } catch {
    console.log("Error:", error.message);
  }
}
```

Figure 1

Identify the THREE errors, then rewrite the complete corrected function.

(5 marks)

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c) State TWO differences between XMLHttpRequest and the Fetch API.

(2 marks)

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QUESTION 3 **[12 MARKS]**

Vue.js Frontend Framework

a) Vue provides `methods`, `computed` properties and `watchers`. State the purpose of EACH and give ONE situation where each one is the right choice.

(3 marks)

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b) Differentiate `ref()` and `reactive()` for declaring reactive state, and state the rule of thumb taught in this course for choosing between them.

(3 marks)

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c) In a Vue Router application, explain the role of EACH of the following: `<router-link>`, `<router-view>`, and a dynamic route parameter such as `/todos/:id` (include how the component reads the parameter's value).

(3 marks)

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d) A user object must be read by components four layers deep in the component tree. Name the problem that passing props through every layer creates, and explain how a *Pinia store* solves it — including the THREE kinds of members a store contains.

(3 marks)

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QUESTION 4 **[15 MARKS]**

Backend, Security and Deployment

a) Each URI below violates RESTful naming conventions. Rewrite EACH as a proper RESTful request (HTTP verb + URI):

i) GET /getAllProducts

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ii) POST /updateProduct/12

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iii) GET /deleteProduct/12

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(3 marks)

b) Name the THREE parts of a JSON Web Token, state what each part contains, and explain why the payload can be read by anyone yet the token is still tamper-proof.

(4 marks)

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c) Differentiate *stored XSS* and *reflected XSS*, and state TWO prevention techniques taught in this course (one on the PHP side and one on the Vue side).

(4 marks)

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d) A team deploys their Vue + Slim project to production. State TWO configuration differences between the development and production environments, and explain why the mobile (Capacitor) build must be created AFTER setting `VITE_API_BASE_URL` to the production URL.

(4 marks)

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SECTION C: CODE APPLICATION QUESTIONS [35 MARKS]

(INSTRUCTION: This section consists of THREE (3) questions. Answer ALL questions. Write syntactically correct code in the spaces provided.)

QUESTION 5 **[10 MARKS]**

HTML Forms, CSS Flexbox and the DOM

a) Write an HTML form fragment for a "Course Registration" section: a `<fieldset>` with the `<legend>` "Student Details", containing an email input that is `required` and properly linked to its `<label>` using `for`/`id`, and a submit button.

(3 marks)

b) Write the CSS so that the element `<nav class="main-nav">` lays out its links in ONE horizontal row with the space distributed BETWEEN the items, and stays fixed at the top of the viewport while the page scrolls (sticky).

(3 marks)

c) Figure 2 shows a list of notes. Each delete button carries the id of its note.

```
<ul id="notesList">
  <li>Buy milk <button class="del" data-id="1">Delete</button></li>
  <li>Call Ali <button class="del" data-id="2">Delete</button></li>
```

```
</ul>
```

Figure 2

Using **EVENT DELEGATION** (ONE listener on the `<ul>` only), write JavaScript so that clicking any delete button: shows `confirm("Are you sure?")` and stops if the user cancels; otherwise removes that button's parent `<li>` from the page. The code must also work for notes added later.

(4 marks)

QUESTION 6 **[10 MARKS]**

Vue Component

Write a COMPLETE Vue 3 Single File Component named `FeedbackBoard.vue` using `<script setup>` that satisfies ALL of the following requirements:

i) receives a prop named `boardTitle` (String, required) and displays it inside an `<h2>`;

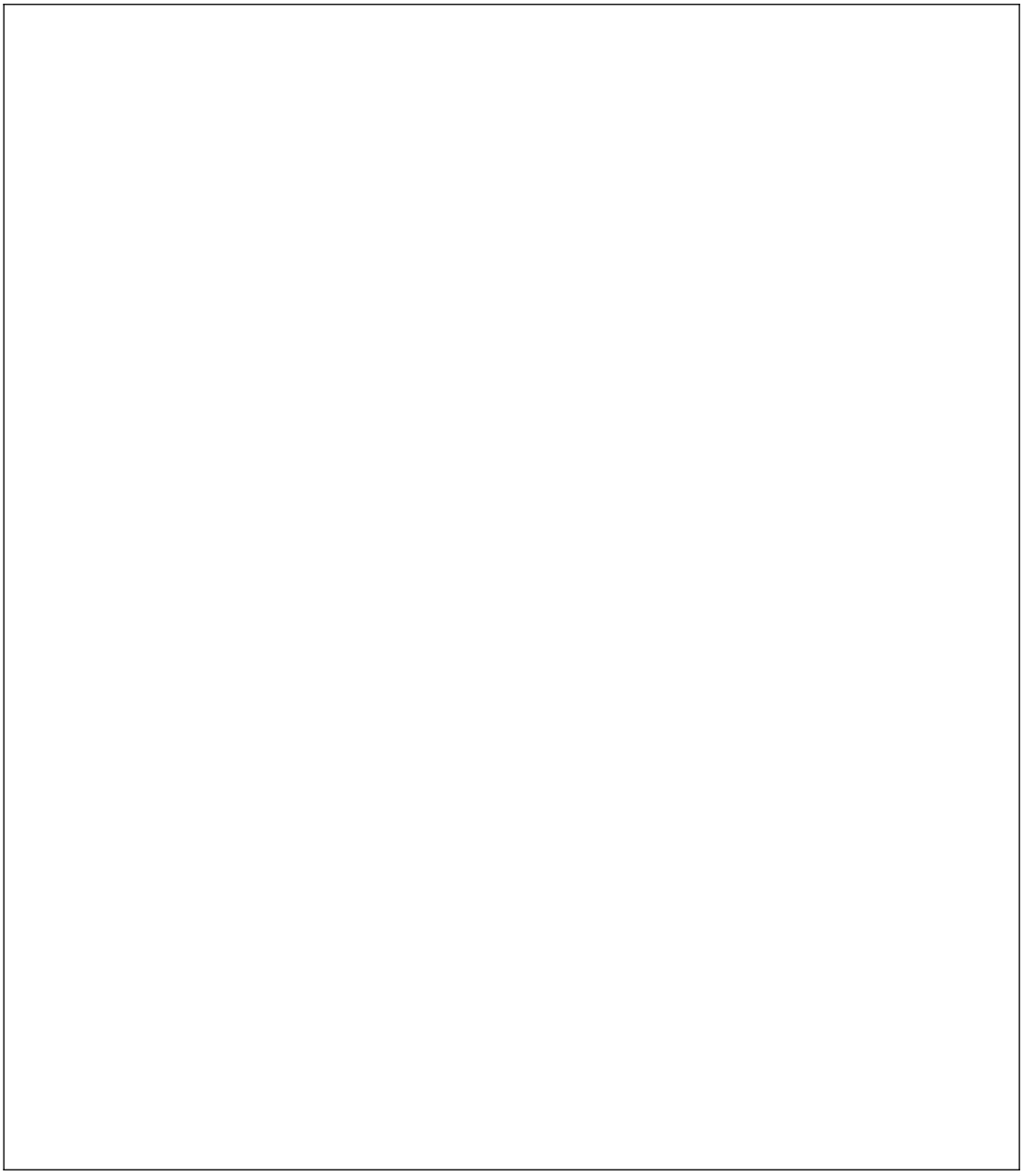
ii) declares a reactive string `comment` bound to a text input with `v-model`;

iii) declares a reactive array `comments`; clicking a "Post" button appends an object `{ id: Date.now(), text: … }` containing the trimmed comment, then clears the input — an empty comment must NOT be posted;

iv) renders the comments with `v-for` using `:key` bound to each comment's `id`, each item showing its `text` and a "Remove" button that deletes that comment from the array;

v) displays the total number of comments using a `computed` property.

(10 marks)



QUESTION 7 **[15 MARKS]**

Full-Stack Backend: Slim 4, PDO and JWT

The *Books API* from your lab work is being extended. Assume the Slim 4 application object `$app` exists and `Request`/`Response` are imported.

a) Write the complete Slim 4 route for `GET /api/books/{id}`. The route must: read the route parameter and cast it to an integer; call the provided function `findBook($id)`, which returns the book as an associative array, or `null` if it does not exist; if the book is not found, return status **404** with the JSON body `{"error":"Book not found"}`; otherwise return status **200** with the book encoded as JSON. Every response must carry `Content-Type: application/json`.

(5 marks)

b) Write PHP code that updates the row with id `$id` in the table `books(title, author, year)` using a PDO **PREPARED STATEMENT** with NAMED placeholders, setting all three columns from `$title`, `$author`, `$year`. After executing, store `true` in `$updated` if at least one row was affected, otherwise `false`.

(5 marks)

c) A login endpoint has fetched the user's record into `$row` (containing `id` and `password_hash`) and the server secret is in `$secret`. Write PHP that: verifies the submitted `$password` against the stored hash using the correct built-in function; and if it is valid, builds a payload containing `sub` (the user id) and `exp` (one hour from now), then encodes it into `$jwt` using the *firebase/php-jwt* library with the `HS256` algorithm.

(3 marks)

d) An attacker intercepts a JWT, edits the payload to change the role to admin, and sends the modified token to the API. Explain how the server DETECTS this tampering, and state the HTTP status code it returns.

(2 marks)

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END OF QUESTION PAPER